SSC MTS 2024 question paper with answers

11 November 2024 · Shift 1

74 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 16 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 13 questions

Q.2

A sum of ₹717 becomes ₹874.74 in 5 years at a certain rate of simple interest. If the rate of interest is increased by 3.1%, then what will be the simple interest (in ₹) of ₹717 in 5 years?

(A) 279.125
(B) 268.875
(C) 211.115
(D) 201.135

Q.5

The average of 11 numbers is 17. If the average of the first five numbers is 16 and the average of the last five numbers is 19, then the sixth number is:

(A) 15
(B) 12
(C) 17
(D) 14

Q.6

Find the HCF of 10, 35 and 55.

(A) 10
(B) 15
(C) 25
(D) 5

Q.7

The average of 5 consecutive natural numbers is 7. Which is the largest number among these?

(A) 8
(B) 10
(C) 7
(D) 9

Q.8

A thief saw a policeman from a distance of 300 m and started running. At the same time, the policeman started chasing the thief. If the speed of the thief is 12 km/h and that of the policeman 15 km/h, then the distance (in km) covered by the policeman to catch the thief is:

(A) 2.0
(B) 2.5
(C) 1.5
(D) 1.0

Q.9

What sum (in ₹) will earn a simple interest of ₹360 in 3 years at 16% per year rate of interest?

(A) 375
(B) 1500
(C) 1000
(D) 750

Q.10

There are 20 cricket balls in the showroom. Among these 20 balls, some are yellow and some are white. The average cost of all the balls is ₹75. If the average cost of yellow balls is ₹80 and that of white is ₹60, then the number of yellow balls is:

(A) 5 **(B)** 6

(C) 14 **(D)** 15

Q.11

The average of 10 numbers is 29. The average of the first 6 numbers is 22 and that of the last 6 numbers is 32. What is the average of the 5th number and the 6th number?

(A) 27 **(B)** 24

(C) 20 **(D)** 17

Q.13

The difference between successive discounts of 25% followed by 12% and 20% followed by 15% on the marked price of a saree is ₹50. The marked price of the saree is:

(A) ₹625 **(B)** ₹5,000

(C) ₹1,250 **(D)** ₹2,500

Q.15

A, B and C start running around a circular track and complete one round in 16, 20 and 28 seconds, respectively. In how much time will they meet again at the starting point?

(A) 310 seconds **(B)** 220 seconds

(C) 510 seconds **(D)** 560 seconds

Q.17

The rate of oil is increased by 12% every year. The rate in the next year would be ₹125.44 per litre. What is the difference between the prices (in ₹) per litre of the oil in the last year and this year?

(A) 15.5 **(B)** 17.5

(C) 14.5 **(D)** 12

Q.18

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 6% per year rate of interest for 3 years.

(A) 480 **(B)** 540

(C) 580 **(D)** 500

Q.19

Find the total surface area of a cone having base radius 49 cm and slant height 61 cm.

(A) 15,900 cm^2 **(B)** 16,940 cm^2

(C) 15,940 cm^2 **(D)** 16,040 cm^2

Reasoning Ability and Problem Solving 14 questions

Q.1

In a certain code language, ‘FROM’ is coded as ‘7519’ and ‘MADE’ is coded as ‘2146’.What is the code for ‘M’ in the given code language?

(A) 2 **(B)** 7

(C) 9 **(D)** 1

Q.3

Seven friends - L, M, N, O, D, E, and F - are seated around a circular table, facing the centre (but not necessarily in the same order). O is sitting fifth to the right of L. M has F and O as immediate neighbours. N and L are immediate neighbours of E. Who is sitting to the immediate left of D?

(A) L
(B) O
(C) M
(D) F

Q.4

In a certain code language, ‘SLAM’ is coded as ‘4918’ and ‘LAME’ is coded as ‘2184’. What is the code for ‘S’ in the given language?

(A) 9
(B) 8
(C) 4
(D) 1

Q.6

Which two numbers should be interchanged to make the following equation correct? (54 ÷ 18)× 5 + 9 × 2 –45 + 12 × 3 = 57 (Note: Interchange should be done of the entire number and not individual digits of a given number.)

(A) 5 and 3
(B) 2 and 5
(C) 18 and 9
(D) 9 and 12

Q.7

When the signs + and ÷ as well as the numbers 5 and 10 are interchanged, which option will be the correct one?

(A) 6 ÷ 10 + 5 = 6
(B) 5 ÷ 1 + 10 = 1
(C) 5 + 10 ÷ 2 = 4
(D) 3 + 5 ÷ 10 = 2

Q.8

In a code language, 'PORTER' is written as 'RVYALP' and 'REMOTE' is written as 'ELTVAR'. How will 'CLOUDY' be written in the same language?

(A) XTVBKC
(B) XSVBKC
(C) YTVBKC
(D) YSVBKC

Q.9

JHMK is related to PNSQ in a certain way based on the English alphabetical order. In the same way, NLQO is related to TRWU . To which of the following is QOTR related, following the same logic?

(A) WUZX
(B) UWXZ
(C) UWZX
(D) WUXZ

Q.11

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g., 13 – Operations on 13 such as adding / subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (15, 14, 13) (12, 10, 8)

(A) (13, 11, 7)
(B) (10, 12, 8)
(C) (15, 15, 15)
(D) (11, 22, 12)

Q.13

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: All shirts are pants. Some shirts are watches. Conclusions: (I) Some pants are watches. (II) Some watches are not shirts.

(A) Only conclusion I follows.
(B) Only conclusion II follows.
(C) Neither conclusion I nor II follows.
(D) Both conclusions I and II follow.

Q.14

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order). E sits second to the right of A. B sits fourth to the right of D. F is an immediate neighbour of E and G. Who sits third to the right of C?

(A) G
(B) E
(C) D
(D) F

Q.17

What should come in place of the question mark (?) in the given series based on the English alphabetical order? HCJ, JGL, LKN, ? , PSR

(A) OOP
(B) NQP
(C) NOP
(D) MST

Q.18

4 is related to 34 following a certain logic. Following the same logic, 7 is related to 61. To which of the following is 11 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 88
(B) 99
(C) 98
(D) 97

Q.19

What should come in place of the question mark (?) in the given series? 28, 31, 36, 43, 54, ?

(A) 66
(B) 65
(C) 64
(D) 67

Q.20

DA2 is related to GD10 in a certain way. In the same way, SP7 is related to VS35. To which of the following is JG10 related following the same logic?

(A) MJ70
(B) GF50
(C) GF70
(D) MJ50

General Awareness 24 questions

Q.1

Drains exiting from Mohenjo-Daro had _____ doors that were probably closed at night to prevent vagrants or negative elements from entering the city through that access.

(A) mud
(B) iron
(C) sandstone
(D) wooden

Q.2
Which of the following was the capital of Vatsa Mahajanapada?

(A) Ujjayini
(B) Kausambi
(C) Ahichchatra
(D) Vaishali

Q.3
In Kerala and coastal area of Karnataka the pre-monsoon shower is known as _____.

(A) Blossom shower
(B) Kalbaisakhi
(C) Mango shower
(D) Bardoisila

Q.4
Former Indian actress Vyjayanthimala was trained in which classical dance form?

(A) Manipuri
(B) Sattriya
(C) Odissi
(D) Bharatanatyam

Q.5
Which of the following is NOT a commercial bank?

(A) Development Bank of Singapore
(B) National Bank for Agriculture and Rural Development
(C) State Bank of India
(D) The Industrial Credit and Investment Corporation of India Bank

Q.6
According to the Human Development Index 2023-24, what is the rank of India?

(A) 134
(B) 132
(C) 130
(D) 129

Q.7
The institution like Lokpal or Lokayukta (Ombudsman) was first created in which of the following countries?

(A) Britain
(B) New Zealand
(C) Norway
(D) Sweden

Q.8
According to Article ____, no religious instruction shall be provided in any educational institution wholly maintained out of State funds.

(A) 28
(B) 27
(C) 29
(D) 30

Q.9
In which year was Pt. Deen Dayal Upadhyaya Grameen Kaushalya Yojana (Scheme) announced by the Government of India?

(A) 2010
(B) 2014
(C) 2012
(D) 2013

Q.10
In june 2023, which state government among the following has announced the 'Shakti' scheme for free bus travel for women and trans people within the state?

(A) Telangana
(B) Tamil Nadu
(C) Kerala
(D) Karnataka

Q.11
Bhamidipati Srivalli Rashmika is associated with which game?

(A) Kho-Kho
(B) Football
(C) Lawn Tennis
(D) Badminton

Q.12
In which weight category did Nitu Ghanghas win the IBA Women's World Boxing Championship event, 2023?

(A) 50 kg
(B) 48 kg
(C) 81 kg
(D) 75 kg

Q.14
Who among the following received the Nritya Kalanidhi for the year 2022?

(A) Bragha Bessell
(B) Geeta Chandran
(C) Kalyanasundaram Pillai
(D) Chitra Visweswaran

Q.15
Vitamin D is synthesised using sunshine, by the:

(A) skin
(B) blood veins
(C) leg
(D) mouth

Q.16
Which festival is celebrated at Sabarimala temple, on the occassion of Makar Sakranti?

(A) Vishu
(B) Onam
(C) Makaravilakku
(D) Ugadi

Q.17
Which structure within the nucleus contains genetic information in the form of DNA?

(A) Nucleolus
(B) Chromosome
(C) Ribosome
(D) Vacuole

Q.18
What is the title of Sania Mirza's autobiography?

(A) Wings of Fire
(B) Changing India
(C) The Test of My Life
(D) Ace Against Odds

Q.19
As on 31st October 2023, the youngest member of parliament (in terms of age) in the 17th Lok Sabha, Chandrani Murmu, represents which political party?

(A) Bharatiya Janata Party
(B) Indian National Congress
(C) Biju Janata Dal
(D) Aam Admi Party

Q.20
Plutonic rocks are formed when lava cools _____.

(A) in the core
(B) in the crust
(C) at the surface
(D) in the asthenosphere

Q.21
Who among the following was the first sultan of Delhi Sultanate to visit the Ajmer shrine of Khwaja Muinuddin Chishti?

(A) Muhammad bin Tughlaq
(B) Jalaluddin Khalji
(C) Ghiyasuddin Balban
(D) Khizr Khan

Q.22
Where was the Hindustan Socialist Republican Association established in the year 1928?

(A) Kanpur
(B) Delhi
(C) Agra
(D) Lahore

Q.23
In which of the following years was a Hindu College in Benaras established to impart education from the ancient Sanskrit texts?

(A) 1790
(B) 1794
(C) 1801
(D) 1791

Q.24
In 2017, who became the first Indian singer to make it to Delhi's Madame Tussauds Museum?

(A) Shreya Ghoshal
(B) Kavita Krishnamurthy
(C) Alisha Chinai
(D) Sunidhi Chauhan

Q.25
Which of the following is the first planned industrial city in India?

(A) Chandigarh
(B) Indore
(C) Navi Mumbai
(D) Jamshedpur

English Language and Comprehension 23 questions

Q.1
Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select 'No substitution'. "This was not only exciting but rewarding" because I was able to experience adventure for the first time in my life.

(A) This was not only excited but also rewarding
(B) No substitution
(C) This was not only exciting but also rewarded
(D) This was not only exciting but also rewarding

Q.2
Select the correctly spelt word to fill in the blank. Macbeth believed the witches' _____ about his future.

(A) prophecy
(B) questionaire
(C) suprise
(D) tyrany

Q.4
Select the appropriate option that can substitute the underlined word in the following sentence to make it grammatically correct. She is the most active and "the more" energetic social worker in our team.

(A) most
(B) the most
(C) the much
(D) very

Q.5
Select the most appropriate option that can substitute the underlined words in the given sentence. We rely on Hiren because he is "most intelligent" than many other boys in our school quiz team.

(A) more intelligent
(B) less intelligent
(C) only intelligent
(D) least intelligent

Q.6

Select the most appropriate ANTONYM of the given word. Boisterous

(A) Bossy
(B) Noisy
(C) Quiet
(D) Cheerful

Q.7

Select the most appropriate option that can substitute the underlined segment in the given sentence. The Lal Chowk is a large open square surrounded by mansions of the “wealthiest” classes than educated ones.

(A) wealthier
(B) wealthy
(C) the wealthy
(D) more wealthier

Q.8

Select the most appropriate synonym of the given word. Insist

(A) Urge
(B) Rebut
(C) Wage
(D) Barge

Q.9

Select the word segment that can substitute the bracketed words correctly and complete the sentence meaningfully. (My husband and I) are from Varanasi.

(A) Everyone of us
(B) Either of us
(C) Both of us
(D) Each one of us

Q.11

Select the INCORRECTLY spelt word.

(A) succlucent
(B) sufficient
(C) sanguinity
(D) subterfuge

Q.12

Select the most appropriate option to fill in the blank. The teacher _____ the students to think critically about the topic.

(A) advised
(B) acquiesced
(C) admonished
(D) assented

Q.13

The following sentence has been split into four segments. Identify the segment that contains a grammatical/spelling error. She has taken / nothing from yesterday / and is feeling / nauseatic now.

(A) nothing from yesterday
(B) and is feeling
(C) She has taken
(D) nauseatic now

Q.14

Select the most appropriate ANTONYM of the underlined word. The strongest suit of documentary theatre lies not only in political insight and “ignorance”, but also in the variety of aesthetic approaches it makes possible.

(A) Mistake
(B) Discrimination
(C) Perspicuity
(D) Acumen

Q.15

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don’t find any error, mark ‘No error’ as your answer. She often goes for a walk / with her dog / in the night.

(A) with her dog
(B) No error
(C) in the night
(D) She often goes for a walk

Q.16

Select the most appropriate synonym of the underlined word. Army officers are given training to “decipher” the secret messages sent during war time.

(A) Decide
(B) Defy
(C) Decode
(D) Encode

Q.17

Select the most appropriate synonym of the given word. Excessive

(A) Exorbitant
(B) Dirty
(C) Scarce
(D) Cute

Q.18

Select the correctly spelled sentence.

(A) The gardner planted colorfull flowers in the garden.
(B) The gardner planted colorful flowers in the gardin.
(C) The gardener planted colourfull flowers in the gardin.
(D) The gardener planted colourful flowers in the garden.

Q.19

Select the most appropriate option to fill in the blank and complete the given phrase. She was on cloud _____ after receiving the scholarship.

(A) ten
(B) eight
(C) seven
(D) nine

Q.20

Select the option that has used the correct tense.

(A) He realised he had forgotten his keys after he has leave the house.
(B) He realised he had forgotten his keys after he had leave the house.
(C) He realised he had forgotten his keys after he was left the house.
(D) He realised he had forgotten his keys after he had left the house.

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As 2023 closes as the warmest year on (1)_____, there is growing global concern about the rise of extreme weather (2)_____ such as floods, droughts and heatwaves. The atmosphere and oceans are warming due to increasing (3)_____ of greenhouse gases, largely caused by burning fossil fuels. This has led to a greater risk of (4)_____ rainfall and flooding. For every 1°C rise in average global temperature, the atmosphere can hold an extra 7% water vapour. When the right atmospheric 'triggers' are in place, this extra water vapour is (5)_____ as intense rainfall.

Select the most appropriate option to fill in blank no. 1.

(A) record
(B) blankness
(C) absence
(D) erasure

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As 2023 closes as the warmest year on (1)_____, there is growing global concern about the rise of extreme weather (2)_____ such as floods, droughts and heatwaves. The atmosphere and oceans are warming due to increasing (3)_____ of greenhouse gases, largely caused by burning fossil fuels. This has led to a greater risk of (4)_____ rainfall and flooding. For every 1°C rise in average global temperature, the atmosphere can hold an extra 7% water vapour. When the right atmospheric 'triggers' are in place, this extra water vapour is (5)_____ as intense rainfall.

Select the most appropriate option to fill in blank no. 2.

(A) quietude
(B) idleness
(C) events
(D) inertness

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As 2023 closes as the warmest year on (1)_____, there is growing global concern about the rise of extreme weather (2)_____ such as floods, droughts and heatwaves. The atmosphere and oceans are warming due to increasing (3)_____ of greenhouse gases, largely caused by burning fossil fuels. This has led to a greater risk of (4)_____ rainfall and flooding. For every 1°C rise in average global temperature, the atmosphere can hold an extra 7% water vapour. When the right atmospheric 'triggers' are in place, this extra water vapour is (5)_____ as intense rainfall.

Select the most appropriate option to fill in blank no. 3.

(A) retentions
(B) immersions
(C) emissions
(D) collections

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As 2023 closes as the warmest year on (1)_____, there is growing global concern about the rise of extreme weather (2)_____ such as floods, droughts and heatwaves. The atmosphere and oceans are warming due to increasing (3)_____ of greenhouse gases, largely caused by burning fossil fuels. This has led to a greater risk of (4)_____ rainfall and flooding. For every 1°C rise inaverage global temperature, the atmosphere can hold an extra 7% water vapour. When the right atmospheric 'triggers' are in place, this extra water vapour is (5)_____ as intense rainfall.

Select the most appropriate option to fill in blank no. 4.

(A) extreme
(B) average
(C) moderate
(D) gentle

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. As 2023 closes as the warmest year on (1)_____, there is growing global concern about the rise of extreme weather (2)_____ such as floods, droughts and heatwaves. The atmosphere and oceans are warming due to increasing (3)_____ of greenhouse gases, largely caused by burning fossil fuels. This has led to a greater risk of (4)_____ rainfall and flooding. For every 1°C rise in average global temperature, the atmosphere can hold an extra 7% water vapour. When the right atmospheric 'triggers' are in place, this extra water vapour is (5)_____ as intense rainfall.

Select the most appropriate option to fill in blank no. 5.

(A) released
(B) confiscated
(C) detained
(D) ensnared

Answer key

Numerical and Mathematical Ability

Q.2 **(B)**	Q.5 **(B)**	Q.6 **(D)**	Q.7 **(D)**	Q.8 **(C)**	Q.9 **(D)**
Q.10 **(D)**	Q.11 **(D)**	Q.13 **(D)**	Q.15 **(D)**	Q.17 **(D)**	Q.18 **(B)**
Q.19 **(B)**					

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.3 **(B)**	Q.4 **(A)**	Q.6 **(C)**	Q.7 **(C)**	Q.8 **(D)**
Q.9 **(A)**	Q.11 **(C)**	Q.13 **(A)**	Q.14 **(B)**	Q.17 **(C)**	Q.18 **(D)**
Q.19 **(D)**	Q.20 **(D)**				

General Awareness

Q.1 **(D)**	Q.2 **(B)**	Q.3 **(C)**	Q.4 **(D)**	Q.5 **(B)**	Q.6 **(A)**
Q.7 **(D)**	Q.8 **(A)**	Q.9 **(B)**	Q.10 **(D)**	Q.11 **(C)**	Q.12 **(B)**
Q.14 **(A)**	Q.15 **(A)**	Q.16 **(C)**	Q.17 **(B)**	Q.18 **(D)**	Q.19 **(C)**
Q.20 **(B)**	Q.21 **(A)**	Q.22 **(B)**	Q.23 **(D)**	Q.24 **(A)**	Q.25 **(D)**

English Language and Comprehension

Q.1 **(D)**	Q.2 **(A)**	Q.4 **(B)**	Q.5 **(A)**	Q.6 **(C)**	Q.7 **(A)**
Q.8 **(A)**	Q.9 **(C)**	Q.11 **(A)**	Q.12 **(A)**	Q.13 **(A)**	Q.14 **(D)**
Q.15 **(C)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(D)**	Q.19 **(D)**	Q.20 **(D)**
Q.21 **(A)**	Q.22 **(C)**	Q.23 **(C)**	Q.24 **(A)**	Q.25 **(A)**	

Explanations

Numerical and Mathematical Ability

Q.2 · Answer (B) 268.875

SI = 874.74 − 717 = ₹157.74, so the rate is 157.74 × 100/(717 × 5) = 4.4%. Raised by 3.1 points to 7.5%, the interest is 717 × 7.5 × 5/100 = ₹268.875.

Q.5 · Answer (B) 12

The 11 numbers total 11 × 17 = 187, the first five 5 × 16 = 80 and the last five 5 × 19 = 95, so the sixth is 187 − 80 − 95 = 12.

Q.6 · Answer (D) 5

10 = 2 × 5, 35 = 5 × 7 and 55 = 5 × 11; the only common factor is 5.

Q.7 · Answer (D) 9

Five consecutive numbers averaging 7 are 5, 6, 7, 8 and 9, so the largest is 9.

Q.8 · Answer (C) 1.5

The policeman gains 15 − 12 = 3 km/h on the 0.3 km gap, so he catches the thief in 0.3/3 = 0.1 hour, having run 15 × 0.1 = 1.5 km.

Q.9 · Answer (D) 750

Principal = SI × 100/(R × T) = 360 × 100/(16 × 3) = ₹750.

Q.10 · Answer (D) 15

With y yellow balls: 80y + 60(20 − y) = 20 × 75 = 1,500, so 20y = 300 and y = 15.

Q.11 · Answer (D) 17

The 10 numbers total 290, the first six 6 × 22 = 132 and the last six 6 × 32 = 192. The 5th and 6th numbers are counted in both, so they total 132 + 192 − 290 = 34 and average 17.

Q.13 · Answer (D) ₹2,500

25% then 12% leaves 0.75 × 0.88 = 66% of the marked price (34% off); 20% then 15% leaves 0.8 × 0.85 = 68% (32% off). The 2% difference is ₹50, so the marked price is ₹2,500.

Q.15 · Answer (D) 560 seconds

They are next together at the start after the LCM of 16, 20 and 28: $2^4 \times 5 \times 7 = 560$ seconds.

Q.17 · Answer (D) 12

This year's rate is 125.44 ÷ 1.12 = ₹112 and last year's 112 ÷ 1.12 = ₹100, so the difference is ₹12.

Q.18 · Answer (B) 540

SI = 3,000 × 6 × 3/100 = ₹540.

Q.19 · Answer (B) 16,940 cm²

TSA = πr(l + r) = 22/7 × 49 × (61 + 49) = 154 × 110 = 16,940 cm².

Reasoning Ability and Problem Solving

Q.1 · Answer (D) 1

M is the only letter in both words and 1 the only digit in both codes, so M = 1.

Q.3 · Answer (B) O

Facing the centre, a person's right is anticlockwise and left clockwise. Clockwise from L the seats are L, D, O, M, F, N, E: O is fifth to L's right, E sits between L and N, M between F and O, and D takes the seat left between L and O. D's immediate left is O.

Q.4 · Answer (A) 9

L, A and M are in both words, as are 1, 4 and 8 in both codes. The letter left in SLAM is S and the digit left in its code is 9, so S = 9.

Q.6 · Answer (C) 18 and 9
Interchanging 18 and 9: (54 ÷ 9) × 5 + 18 × 2 − 45 + 12 × 3 = 30 + 36 − 45 + 36 = 57.

Q.7 · Answer (C) 5 + 10 ÷ 2 = 4
After the interchanges 5 + 10 ÷ 2 = 4 reads 10 ÷ 5 + 2 = 2 + 2 = 4. The others give 6 + 5 ÷ 10 = 6.5, 10 + 1 ÷ 5 = 10.2 and 3 ÷ 10 + 5 = 5.3.

Q.8 · Answer (D) YSVBKC
The first and last letters swap places and each letter between them moves 7 places forward (T wraps round to A): PORTER → R VYAL P and REMOTE → E LTVA R. So CLOUDY → Y SVBK C: YSVBKC.

Q.9 · Answer (A) WUZX
Each letter moves 6 places forward: J → P, H → N, M → S, K → Q, and N → T, L → R, Q → W, O → U. So QOTR → WUZX.

Q.11 · Answer (C) (15, 15, 15)
In each set the middle number is the average of the other two: 15 + 13 = 2 × 14 and 12 + 8 = 2 × 10. Only (15, 15, 15) fits: 15 + 15 = 2 × 15.

Q.13 · Answer (A) Only conclusion I follows.
The shirts that are watches are also pants, so some pants are watches (I). Every watch could be a shirt, so II need not follow.

Q.14 · Answer (B) E
Facing the centre, a person's right is anticlockwise. Clockwise from A the seats are A, C, B, G, F, E, D: E is second to A's right, F sits between E and G, and only D in the seat on A's right puts B fourth to D's right. Third to the right of C is E.

Q.17 · Answer (C) NOP
First letters move 2 forward (H, J, L, N, P), second letters 4 forward (C, G, K, O, S) and third letters 2 forward (J, L, N, P, R): NOP.

Q.18 · Answer (D) 97
The second number is 9 times the first minus 2: 9 × 4 − 2 = 34 and 9 × 7 − 2 = 61. Likewise 9 × 11 − 2 = 97.

Q.19 · Answer (D) 67
The differences are 3, 5, 7, 11, successive prime numbers, so the next is 13: 54 + 13 = 67.

Q.20 · Answer (D) MJ50
Both letters move 3 places forward and the number is multiplied by 5: DA2 → GD10 and SP7 → VS35. So JG10 → MJ50.

General Awareness

Q.1 · Answer (D) wooden
SSC's key gives wooden doors. The Harappans did not use iron, and a door of mud or sandstone could not easily be opened and shut every night.

Q.2 · Answer (B) Kausambi
Kausambi, on the Yamuna, was the capital of Vatsa. Ujjayini was the capital of Avanti, Vaishali of the Vajji confederacy and Ahichchhatra of northern Panchala.

Q.3 · Answer (C) Mango shower
The pre-monsoon showers of Kerala and coastal Karnataka are called mango showers, as they help mangoes ripen early. Blossom showers make coffee flowers bloom, and Kalbaisakhi (Bardoisila in Assam) are the evening thunderstorms of Bengal and Assam.

Q.4 · Answer (D) Bharatanatyam
Vyjayanthimala trained in Bharatanatyam, which she performed alongside her film career.

Q.5 · Answer (B) National Bank for Agriculture and Rural Development
NABARD, set up in 1982, is an apex development finance institution for agriculture and rural development, not a commercial bank. DBS, SBI and ICICI Bank are commercial banks.

Q.6 · Answer (A) 134
UNDP's Human Development Report 2023-24 ranks India 134th of 193 countries.

Q.7 · Answer (D) Sweden
The ombudsman was first created in Sweden, in 1809; the Lokpal and Lokayuktas are India's version of it.

Q.8 · Answer (A) 28
Article 28(1) bars religious instruction in any educational institution wholly maintained out of State funds.

Q.9 · Answer (B) 2014
DDU-GKY, the Ministry of Rural Development's placement-linked skilling scheme for rural youth, was announced on 25 September 2014, Deen Dayal Upadhyaya's birth anniversary.

Q.10 · Answer (D) Karnataka
Karnataka's Shakti scheme, launched on 11 June 2023, lets women and transgender persons travel free in the state's government buses within Karnataka.

Q.11 · Answer (C) Lawn Tennis
Shrivalli Rashmikaa Bhamidipaty, from Hyderabad, is a tennis player.

Q.12 · Answer (B) 48 kg
Nitu Ghanghas won gold in the 48 kg category at the 2023 IBA Women's World Boxing Championships in New Delhi.

Q.14 · Answer (A) Bragha Bessell
The Music Academy, Madras, named the Bharatanatyam dancer and guru Bragha Bessell its Nritya Kalanidhi for 2022.

Q.15 · Answer (A) skin
The skin makes vitamin D when it is exposed to the ultraviolet rays in sunlight.

Q.16 · Answer (C) Makaravilakku
Makaravilakku is the festival held at Sabarimala on Makar Sankranti. Vishu is the Malayali new year, Onam Kerala's harvest festival and Ugadi the Telugu and Kannada new year.

Q.17 · Answer (B) Chromosome
Chromosomes, made of DNA and proteins, carry the genetic information. The nucleolus makes ribosomes, which build proteins in the cytoplasm.

Q.18 · Answer (D) Ace Against Odds
Sania Mirza's autobiography is 'Ace Against Odds' (2016). 'Wings of Fire' is A. P. J. Abdul Kalam's, 'The Test of My Life' Yuvraj Singh's and 'Changing India' Manmohan Singh's.

Q.19 · Answer (C) Biju Janata Dal
Chandrani Murmu, elected from Keonjhar, Odisha, in 2019 at the age of 25, was the youngest member of the 17th Lok Sabha; she belongs to the Biju Janata Dal.

Q.20 · Answer (B) in the crust
Plutonic (intrusive) igneous rocks such as granite form when molten rock cools slowly deep inside the crust, where it is strictly called magma. Lava that cools at the surface forms volcanic rocks such as basalt.

Q.21 · Answer (A) Muhammad bin Tughlaq
Muhammad bin Tughlaq (ruled 1324-51) was the first Sultan to visit the shrine of Khwaja Muinuddin Chishti at Ajmer.

Q.22 · Answer (B) Delhi
The Hindustan Socialist Republican Association was founded in September 1928 at Feroz Shah Kotla in Delhi, by Chandrashekhar Azad, Bhagat Singh and others.

Q.23 · Answer (D) 1791
Jonathan Duncan set up the Hindu College at Benaras in 1791 to encourage the study of ancient Sanskrit texts; the Calcutta Madrasa had been set up in 1781.

Q.24 · Answer (A) Shreya Ghoshal
In 2017 Shreya Ghoshal became the first Indian singer to have a wax figure at Madame Tussauds, Delhi.

Q.25 · Answer (D) Jamshedpur
Jamshedpur, built around the Tata iron and steel works founded in 1907, is regarded as India's first planned industrial city.

English Language and Comprehension

Q.1 · Answer (D) This was not only exciting but also rewarding
'Not only' pairs with 'but also': 'not only exciting but also rewarding'.

Q.2 · Answer (A) prophecy
'Prophecy' (a prediction) is spelt correctly and fits; the others should be 'questionnaire', 'surprise' and 'tyranny'.

Q.4 · Answer (B) the most
Both adjectives single her out in the team, so both take the superlative: 'the most active and the most energetic'.

Q.5 · Answer (A) more intelligent
'Than' needs a comparative: 'more intelligent than many other boys'. 'Less intelligent' would give no reason to rely on him.

Q.6 · Answer (C) Quiet
'Boisterous' means noisy and lively; its opposite is 'quiet'.

Q.7 · Answer (A) wealthier
'Than' needs a comparative: 'the wealthier classes than educated ones'. 'More wealthier' is a double comparative.

Q.8 · Answer (A) Urge
To 'insist' is to demand or press for something firmly; 'urge' is closest.

Q.9 · Answer (C) Both of us
Two people with the plural verb 'are' make 'both of us'; 'either of us', 'each one of us' and 'everyone of us' take a singular verb.

Q.11 · Answer (A) succlucent
The correct spelling is 'succulent'.

Q.12 · Answer (A) advised
SSC's key gives 'advised': one advises someone to do something. 'Acquiesced' and 'assented' mean agreed and cannot take 'the students' as an object; 'admonished' can mean 'urged firmly', but it carries a reproof the sentence does not suggest.

Q.13 · Answer (A) nothing from yesterday
SSC's key gives 'nothing from yesterday': with the present perfect, the starting point takes 'since', 'has taken nothing since yesterday'. 'Nauseous' or 'nauseated' is the usual word for 'nauseatic', so the last segment is arguable too.

Q.14 · Answer (D) Acumen
'Ignorance' is lack of knowledge or understanding; its opposite is 'acumen', keen insight and judgement. 'Perspicuity' means clarity of expression.

Q.15 · Answer (C) in the night
Something done regularly after dark is done 'at night': 'goes for a walk with her dog at night'.

Q.16 · Answer (C) Decode
To 'decipher' a message is to 'decode' it.

Q.17 · Answer (A) Exorbitant
'Excessive' means more than is reasonable: 'exorbitant'.

Q.18 · Answer (D) The gardener planted colourful flowers in the garden.
The correct spellings are 'gardener', 'colourful' (with one 'l' at the end) and 'garden'.

Q.19 · Answer (D) nine
To be 'on cloud nine' is to be extremely happy.

Q.20 · Answer (D) He realised he had forgotten his keys after he had left the house.
The earlier of two past actions takes the past perfect, 'had' + past participle: 'after he had left the house'.

Q.21 · Answer (A) record
2023 was the warmest year 'on record', that is since measurements began.

Q.22 · Answer (C) events
Floods, droughts and heatwaves are extreme weather 'events'.

Q.23 · Answer (C) emissions
Burning fossil fuels gives off 'emissions' of greenhouse gases, which warm the atmosphere and oceans.

Q.24 · Answer (A) extreme
A warmer atmosphere holds more water vapour, so the risk is of 'extreme' rainfall.

Q.25 · Answer (A) released
The extra water vapour is 'released' as intense rainfall.

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